

Flow42 v0.8 Export Data Dictionary / Backtesting Handoff

Dataset context

Source file: `Flow42_ReversalFeatures.csv` from `Flow42DataCapture_v0_8`.

Chart/run context:

```
Instrument: NQ JUN26
BarsPeriod: 1 Min Volumetric
Preset: Export_FastAllFlow42
Rows: 10,000
Time range: 2026-06-05 12:25:00 through 2026-06-16 20:06:00
Preferred delta source: NTVolumetricBidAsk
```

Important global interpretation:

```
Use PreferredDelta_* or NTVolumetric_* for true bid/ask delta research.
Do not rely on Flow42 native VolDelta raw Delta/CVD as true bid/ask delta in
this run.
Flow42 native VolDelta appears proxy-like because |Delta| equals full bar volume
on ~98.6% of rows.
Blank columns mean no strict trusted output was found; blanks are not zero.
Draw-object event columns are all zero in this run because Flow42 visual objects
were not captured as NinjaTrader DrawObjects.
```

Highest-confidence usable feature families

Use these first for modeling:

1. OHLCV / candle geometry
2. PreferredDelta_* / NTVolumetric_* true bid/ask delta
3. VolDrive continuation signals
4. ChopState state / chop score
5. VolSpike flag, but note it fires very often in this run

Be careful with these:

Flow42 native VolDelta_* = diagnostic/proxy-like in this run
VolImbalance_Bid / Ask = exposed but all zero in this run
VolSpike_Flag = populated but broad, about 68% of rows

Currently unavailable / placeholder-only from this run:

AbsorptionMap stage/side/price/strength
VolDynamics absorption/exhaustion/push/surge
VolImbalance cluster details
CloudNotes levels
VolLiqMeter live DOM/liquidity values

Variable list and descriptions

1. Run / metadata columns

Column	Description	Usefulness
RunId	Human-readable run folder/name identifier. Constant across the file.	Metadata only.
ExportTime	Time the exporter wrote each row.	Usually not a model feature. Useful for audit.
TriggeredBy	Export trigger, e.g. LOAD_COMPLETE_SCAN.	Metadata only.
Instrument	Instrument name, here NQ JUN26.	Metadata/filter.
BarsPeriod	Chart period, here 1 Min Volumetric.	Metadata/filter.
BarIndex	NinjaTrader bar index.	Useful for alignment/joins.
BarTime	Timestamp of the bar.	Essential for joining with other datasets.

2. OHLCV / candle structure

Column	Description	Usefulness
Open	Bar open price.	Core market data.
High	Bar high price.	Core market data.

Column	Description	Usefulness
<code>Low</code>	Bar low price.	Core market data.
<code>Close</code>	Bar close price.	Core market data.
<code>Volume</code>	Total bar volume.	Core market data.
<code>Body</code>	Absolute candle body size, usually $\text{abs}(\text{Close} - \text{Open})$.	Useful for momentum/rejection/body filters.
<code>Range</code>	Candle range, $\text{High} - \text{Low}$.	Useful for volatility and rejection filters.
<code>UpperWick</code>	Distance from upper body edge to high.	Useful for failed high / rejection detection.
<code>LowerWick</code>	Distance from low to lower body edge.	Useful for failed low / rejection detection.
<code>BodyFrac</code>	Body as fraction of total range.	Useful for doji/stall/rejection filters.
<code>UpperWickFrac</code>	Upper wick as fraction of range.	Useful for short reversal / failed high filters.
<code>LowerWickFrac</code>	Lower wick as fraction of range.	Useful for long reversal / failed low filters.
<code>CloseLocationFrac</code>	Close location within bar range, usually $(\text{Close} - \text{Low}) / \text{Range}$. Near 1 = closed high, near 0 = closed low.	Very useful for rejection/follow-through logic.

3. VolDrive continuation outputs

VolDrive is a continuation/trend-confirmation tool, not a pure reversal detector. Use it as an anti-fade filter or post-reversal confirmation.

Column	Description	Usefulness
<code>VolDrive_Long</code>	Flow42 VolDrive long continuation output. In this run, <code>1</code> on 62 bars, otherwise <code>0</code> .	Useful as bullish continuation confirmation.
<code>VolDrive_Short</code>	Flow42 VolDrive short continuation output. In this run, <code>-1</code> on 58 bars, otherwise <code>0</code> .	Useful as bearish continuation confirmation.
<code>VolDrive_Signal</code>	Combined signed VolDrive signal: <code>1</code> long, <code>-1</code> short, <code>0</code> neutral.	Best VolDrive column to use directly.

Suggested use:

For reversal testing:

- Opposite-direction VolDrive active near entry = caution / anti-fade warning.
- VolDrive flips into reversal direction after entry = confirmation.

4. Flow42 native VolDelta outputs — diagnostic / proxy-like in this run

These are Flow42's own VolDelta outputs. They are exposed cleanly, but in this specific run they appear to be using a proxy/up-down style classification.

Column	Description	Usefulness
<code>VolDelta_BuyVolume</code>	Flow42 native buy volume estimate.	Diagnostic only in this run.
<code>VolDelta_SellVolume</code>	Flow42 native sell volume estimate.	Diagnostic only in this run.
<code>VolDelta_Delta</code>	Flow42 native delta.	Do not treat as true bid/ask delta in this run.
<code>VolDelta_CVD</code>	Flow42 native cumulative delta.	Do not treat as true bid/ask CVD in this run.
<code>VolDelta_RelativeDeltaPressure</code>	Flow42 native relative pressure value, generally between <code>-1</code> and <code>1</code> .	Potentially useful as vendor-derived pressure feature, but validate separately.

Column	Description	Usefulness
<code>VolDelta_SourceOrMode</code>	Combined text showing Flow42 VolDelta mode/settings. Here: <code>DeltaSource=Auto;CvdMode=Cumulative;Reset=Session</code> .	Diagnostic
<code>VolDelta_DeltaSourceSetting</code>	Flow42 DeltaSource setting. Here <code>Auto</code> .	Diagnostic
<code>VolDelta_CvdMode</code>	Flow42 CVD mode. Here <code>Cumulative</code> .	Diagnostic
<code>VolDelta_CvdResetMode</code>	Flow42 CVD reset mode. Here <code>Session</code> .	Diagnostic
<code>VolDelta_AbsDeltaEqualsVolumeFlag</code>	<code>1</code> when Flow42 <code>abs(Delta)</code> equals full bar <code>Volume</code> ; this happened on about 98.6% of rows.	Source-quality warning flag.
<code>VolDelta_BuySellSumMinusVolume</code>	Flow42 buy + sell minus total volume. Here always <code>0</code> , meaning Flow42 buy/sell internally sums to total volume.	Diagnostic

Important warning:

Flow42 native VolDelta in this export behaves like:

- Up bar: nearly all volume assigned to buy
- Down bar: nearly all volume assigned to sell
- Flat bar: split volume

So it should not be used as the primary delta/CVD source for serious order-flow backtesting.

5. NinjaTrader volumetric bid/ask delta — primary true delta source

These columns are exporter-calculated from NinjaTrader's 1-minute volumetric bars. These are the preferred delta features.

Column	Description	Usefulness
<code>NTVolumetric_BuyVolume</code>	Buy/aggressor-side volume from NinjaTrader volumetric bar data.	High. Use for true bid/ask volume.
<code>NTVolumetric_SellVolume</code>	Sell/aggressor-side volume from NinjaTrader volumetric bar data.	High. Use for true bid/ask volume.

Column	Description	Usefulness
<code>NTVolumetric_Delta</code>	<code>BuyVolume</code> - <code>SellVolume</code> from volumetric data.	High. Use as primary bar delta.
<code>NTVolumetric_CVD_Run</code>	Running cumulative delta across the exported 10,000 rows.	High, but note it is export-window cumulative, not necessarily full-session unless export starts at session boundary.
<code>NTVolumetric_Available</code>	<code>1</code> if NT volumetric bid/ask data was successfully available for that row. Here it is <code>1</code> for all rows.	Quality flag.
<code>NTVolumetric_SourceStatus</code>	Text status. Here <code>NT Volumetric BidAsk</code> .	Source audit.
<code>NTVolumetric_BuySellSumMinusVolume</code>	<code>(NT buy + NT sell) - bar volume</code> . Here <code>0</code> for all rows.	Data-integrity check.

Key quality result:

```
NT volumetric delta worked correctly:
- 10,000 / 10,000 rows available
- 0 missing rows
- 0 buy/sell/volume mismatches
- |Delta| equals full volume only ~1.5% of rows, which is much more realistic
```

6. PreferredDelta columns — easiest delta columns to use

These columns choose the best available delta source based on exporter settings. In this run, they use NT volumetric bid/ask on every row.

Column	Description	Usefulness
<code>PreferredDelta_BuyVolume</code>	Preferred buy volume. In this run, same as <code>NTVolumetric_BuyVolume</code> .	High. Use this for modeling.
<code>PreferredDelta_SellVolume</code>	Preferred sell volume. In this run, same as <code>NTVolumetric_SellVolume</code> .	High. Use this for modeling.

Column	Description	Usefulness
<code>PreferredDelta_Delta</code>	Preferred bar delta. In this run, same as <code>NTVolumetric_Delta</code> .	High. Use as primary delta.
<code>PreferredDelta_CVD</code>	Preferred cumulative delta. In this run, same as <code>NTVolumetric_CVD_Run</code> .	High, with export-window/session-boundary caveat.
<code>PreferredDelta_Source</code>	Source used for preferred delta. Here always <code>NTVolumetricBidAsk</code> .	Essential audit column.

Recommended rule:

Use `PreferredDelta_*` by default.
Only use `NTVolumetric_*` directly if you specifically want to force the raw NT volumetric values.
Do not use Flow42 native `VolDelta_*` as the primary delta source in this run.

7. ChopState regime columns

ChopState gives market-regime context: choppy vs directional.

Column	Description	Usefulness
<code>ChopState_State</code>	Flow42 ChopState public output. In this run, <code>1</code> on 6,325 rows and <code>0</code> on 3,675 rows. Likely choppy/active state flag.	Useful regime feature.
<code>ChopState_ChopScore</code>	Flow42 ChopScore output. Values observed: <code>1</code> , <code>0</code> , <code>-1</code> . In this run, mostly <code>1</code> .	Useful as a regime/quality score.
<code>ChopState_ER</code>	Placeholder for efficiency ratio if exactly exposed. Blank in this run.	Not usable currently.
<code>ChopState_Overlap</code>	Placeholder for overlap/chop overlap output if exactly exposed. Blank in this run.	Not usable currently.
<code>ChopState_BoxPositionOrEdge</code>	Placeholder for chop-box position or edge state if exposed. Blank in this run.	Not usable currently.

Suggested use:

Test reversals separately under:
- `ChopState_State = 1`

- ChopState_State = 0
- ChopScore = 1 / 0 / -1

Do not assume chop is always bad. Some reversals may work best at range/chop edges, while continuation trades may prefer non-chop.

8. VolSpike columns

VolSpike flags volume spike behavior. In this export it fires frequently, so it may need tuning.

Column	Description	Usefulness
VolSpike_Flag	VolSpike public/API flag. In this run, 1 on 6,835 rows and 0 on 3,165 rows.	Usable, but broad; not rare enough by itself.
VolSpike_RatioOrZScore	Placeholder for spike ratio/z-score if exposed. Blank in this run.	Not usable currently.
VolSpike_Mode	Placeholder for VolSpike mode/extra context if exposed. Blank in this run.	Not usable currently.

Suggested use:

Do not treat VolSpike_Flag alone as a reversal signal.
Use it as an attention/context flag:

- spike + failed high/low
- spike + large wick
- spike + delta divergence
- spike near prior extreme/support/resistance

9. VolDynamics columns

These are desired VolDynamics event features, but they were not cleanly exposed in this run.

Column	Description	Usefulness
VolDynamics_Absorption	Placeholder for VolDynamics absorption event flag. Blank.	Not usable currently.
VolDynamics_Exhaustion	Placeholder for VolDynamics exhaustion event flag. Blank.	Not usable currently.

Column	Description	Usefulness
<code>VolDynamics_VolumePush</code>	Placeholder for volume push / initiation event. Blank.	Not usable currently.
<code>VolDynamics_VolumeSurge</code>	Placeholder for surge event. Blank.	Not usable currently.

Interpretation:

Deep discovery found some VolDynamics-related settings/baselines, but not strict trustworthy event outputs. These columns were intentionally left blank rather than filled with loose/fake matches.

10. VolImbalance columns

VolImbalance exposes BloodHound-style BID/ASK signals, but they did not fire in this run.

Column	Description	Usefulness
<code>VolImbalance_Bid</code>	Flow42 VolImbalance BloodHound BID imbalance signal. Exists but was <code>0</code> for all rows.	Currently no signal in this window/settings.
<code>VolImbalance_Ask</code>	Flow42 VolImbalance BloodHound ASK imbalance signal. Exists but was <code>0</code> for all rows.	Currently no signal in this window/settings.
<code>VolImbalance_ClusterCount</code>	Placeholder for cluster count if exposed. Blank.	Not usable currently.
<code>VolImbalance_ClusterVolume</code>	Placeholder for cluster volume if exposed. Blank.	Not usable currently.
<code>VolImbalance_ClusterSideOrStrength</code>	Placeholder for cluster side/strength if exposed. Blank.	Not usable currently.

Interpretation:

The basic BID/ASK automation outputs exist, but they produced no active signals in this export. Cluster/zone details were not cleanly exposed.

11. AbsorptionMap columns

These are desired AbsorptionMap event columns, but they were not cleanly exposed in this run.

Column	Description	Usefulness
<code>AbsorptionMap_Stage</code>	Intended stage such as live warning/candidate/provisional/confirmed. Blank.	Not usable currently.
<code>AbsorptionMap_Side</code>	Intended absorbed side/direction. Blank.	Not usable currently.
<code>AbsorptionMap_ScoreOrStrength</code>	Intended absorption strength/score. Blank.	Not usable currently.
<code>AbsorptionMap_Price</code>	Intended absorption price/level. Blank.	Not usable currently.

Interpretation:

Private-field discovery suggests AbsorptionMap likely has richer internal/private structures, but no clean public `Values[]` / `public Series` / trusted export output was found.

12. VolLiqMeter columns

VolLiqMeter is a live DOM/liquidity tool and was not part of the historical export.

Column	Description	Usefulness
<code>VolLiqMeter_AggressorBuy</code>	Placeholder for live aggressor buy volume. Blank.	Not usable in this historical run.
<code>VolLiqMeter_AggressorSell</code>	Placeholder for live aggressor sell volume. Blank.	Not usable in this historical run.
<code>VolLiqMeter_Imbalance</code>	Placeholder for liquidity imbalance. Blank.	Not usable in this historical run.
<code>VolLiqMeter_AvgLiquidity</code>	Placeholder for average liquidity/depth. Blank.	Not usable in this historical run.

Interpretation:

VolligMeter should be tested separately during an active live session with Level 2 data. It should not be assumed reconstructable historically from this export.

13. CloudNotes columns

CloudNotes is only useful if external levels are configured and readable.

Column	Description	Usefulness
CloudNotes_NearestLevelOrDistance	Placeholder for nearest external/manual/gamma/CSV level or distance. Blank.	Not usable currently.
CloudNotes_LevelType	Placeholder for type of nearest level. Blank.	Not usable currently.

Interpretation:

CloudNotes was not meaningfully active/configured in this run. Use a separate CloudNotes-focused run only if levels are configured.

14. Draw-object event columns

These columns are designed to count Flow42 chart markers/draw objects on the same bar or over the last three bars. In this run, they are all zero because no real Flow42 draw objects were captured as NinjaTrader DrawObjects.

Same-bar draw object columns

Column	Description	Usefulness
DrawEvents_SameBar_Total	Total captured draw events on the same bar.	All zero currently.
DrawEvents_SameBar_AbsorptionMap	Same-bar captured AbsorptionMap draw events.	All zero currently.
DrawEvents_SameBar_VolImbalance	Same-bar captured VolImbalance draw events.	All zero currently.
DrawEvents_SameBar_VolDynamics	Same-bar captured VolDynamics draw events.	All zero currently.

Column	Description	Usefulness
<code>DrawEvents_SameBar_VolSpike</code>	Same-bar captured VolSpike draw events.	All zero currently.
<code>DrawEvents_SameBar_VolDrive</code>	Same-bar captured VolDrive draw events.	All zero currently.
<code>DrawEvents_SameBar_VolDelta</code>	Same-bar captured VolDelta draw events.	All zero currently.
<code>DrawEvents_SameBar_ChopState</code>	Same-bar captured ChopState draw events.	All zero currently.
<code>DrawEvents_SameBar_CloudNotes</code>	Same-bar captured CloudNotes draw events.	All zero currently.
<code>DrawEvents_SameBar_VolLiqMeter</code>	Same-bar captured VolLiqMeter draw events.	All zero currently.

Last-three-bar draw object columns

Column	Description	Usefulness
<code>DrawEvents_Last3_Total</code>	Total captured draw events over current + previous 2 bars.	All zero currently.
<code>DrawEvents_Last3_Candidate</code>	Last-3 draw events classified as candidate.	All zero currently.
<code>DrawEvents_Last3_Provisional</code>	Last-3 draw events classified as provisional.	All zero currently.
<code>DrawEvents_Last3_Confirmed</code>	Last-3 draw events classified as confirmed.	All zero currently.
<code>DrawEvents_Last3_LiveWarning</code>	Last-3 draw events classified as live warning.	All zero currently.
<code>DrawEvents_Last3_Absorption</code>	Last-3 draw events classified as absorption.	All zero currently.
<code>DrawEvents_Last3_Exhaustion</code>	Last-3 draw events classified as exhaustion.	All zero currently.
<code>DrawEvents_Last3_Imbalance</code>	Last-3 draw events classified as imbalance.	All zero currently.
<code>DrawEvents_Last3_Spike</code>	Last-3 draw events classified as spike.	All zero currently.
<code>DrawEvents_Last3_DeltaOrCVD</code>	Last-3 draw events classified as delta/CVD.	All zero currently.

Column	Description	Usefulness
<code>DrawEvents_Last3_Up</code>	Last-3 draw events inferred as upward/ bullish.	All zero currently.
<code>DrawEvents_Last3_Down</code>	Last-3 draw events inferred as downward/ bearish.	All zero currently.

Interpretation:

Do not use draw-event columns from this run. They are included for schema completeness, but Flow42 visuals appear not to be captured as standard NinjaTrader DrawObjects in this setup.

Practical modeling recommendations

Use first

```

PreferredDelta_Delta
PreferredDelta_CVD
PreferredDelta_BuyVolume
PreferredDelta_SellVolume
PreferredDelta_Source

NTVolumetric_Delta
NTVolumetric_CVD_Run
NTVolumetric_BuyVolume
NTVolumetric_SellVolume

VolDrive_Signal
VolDrive_Long
VolDrive_Short

ChopState_State
ChopState_ChopScore

VolSpike_Flag

OHLCV / candle-structure columns

```

Use cautiously

```
VolDelta_RelativeDeltaPressure  
VolSpike_Flag  
VolImbalance_Bid  
VolImbalance_Ask
```

Do not use as model features from this run

```
Flow42 native VolDelta_Delta / CVD as true bid/ask delta  
AbsorptionMap_* blank columns  
VolDynamics_* blank columns  
VolImbalance cluster columns  
VolLiqMeter_* columns  
CloudNotes_* columns  
DrawEvents_* columns
```

Main thesis for backtesting threads

This dataset gives us a clean Flow42-compatible feature layer for:

- price/candle structure
- true volumetric bid/ask delta
- CVD/pressure via PreferredDelta
- VolDrive continuation confirmation
- ChopState regime filtering
- VolSpike context

It does not yet give clean public/exportable versions of:

- AbsorptionMap staged absorption events
- VolDynamics absorption/exhaustion/push/surge events
- VolImbalance cluster/zone details
- CloudNotes external levels
- VolLiqMeter live DOM/liquidity context

So the correct use is to treat this as a partial but useful feature set, not a complete Flow42 visual-event scrape.